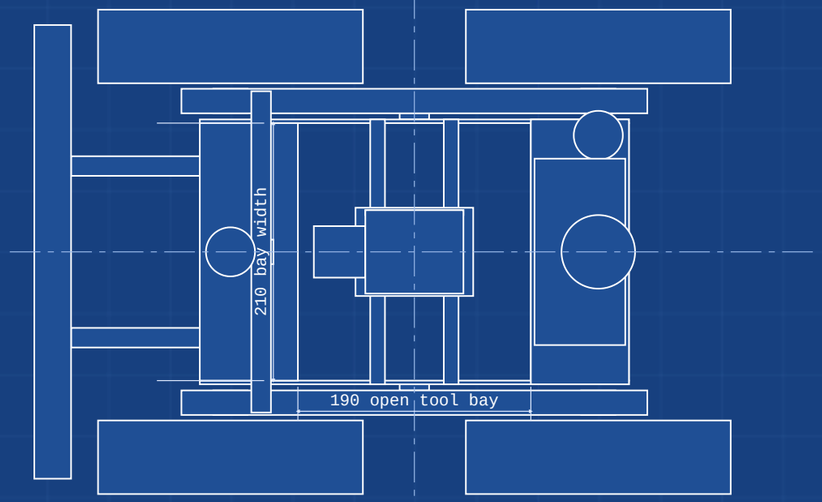
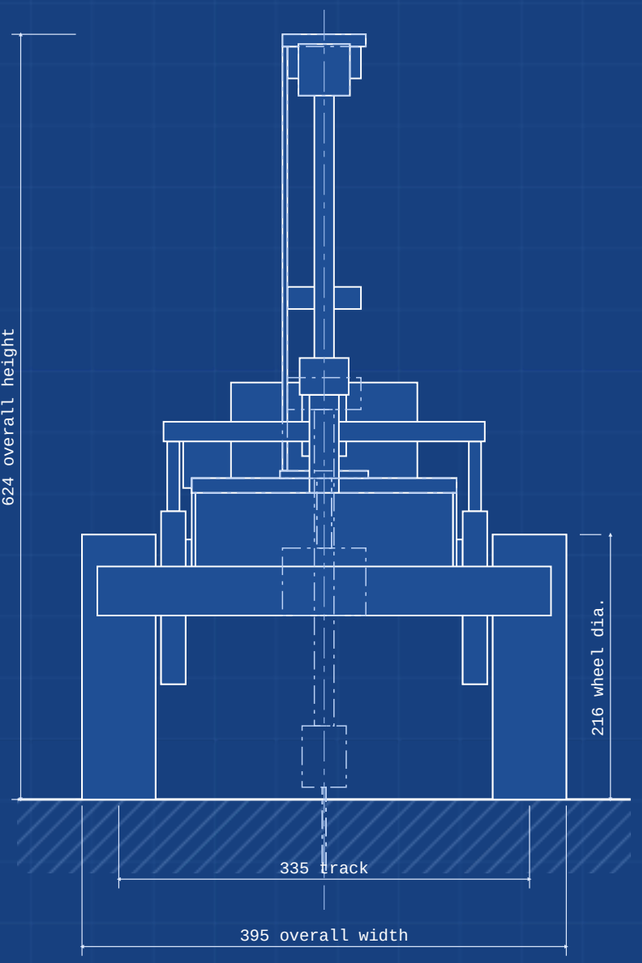
GENERAL ARRANGEMENT

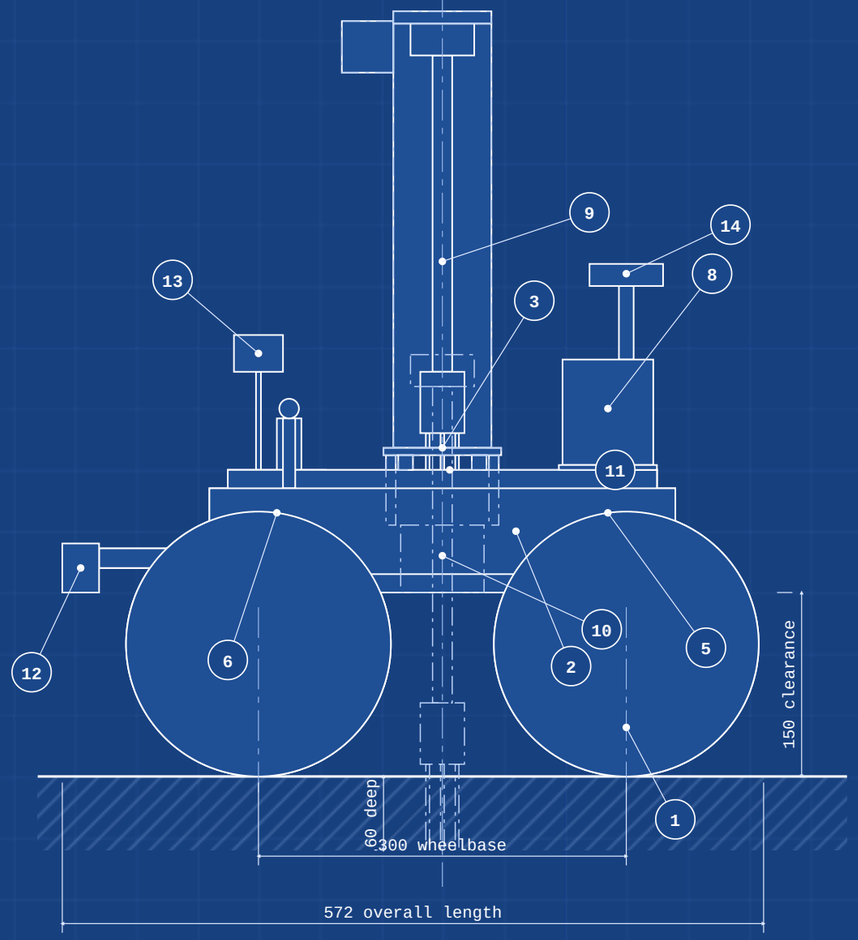
Rocker-differential chassis on four hub-motor wheels.

PLAN





FRONT ELEVATION



LEFT SIDE ELEVATION

PARTS LIST

REF	ITEM
1	Hub-motor wheels 8.5 in pneumatic, BLDC with halls, 4 (sealant-f…
2	Side rocker beams 30 x 70 aluminium box, with axle drop brackets…
3	Rocker pivots (stainless stub shafts, sealed bearings) and diffe…
4	Body side plates, 3 mm aluminium, carrying the rocker pivots; ca…
5	Front cross box: 2 dual-channel BLDC FOC controllers with curren…
6	Rear cross box: 12S LiFePO4 6 Ah pack, BMS, main fuse, e-stop co…
7	Rear deck plate, 3 mm aluminium
8	IP65 electronics enclosure: Pi 5, MCU, DC-DC rails, isolated RS4…
9	Probe module: two Y rails and a sled; belt linear axis behind a …
10	Cleaning station: PU wiper plate, strip brushes, air/water ring;…
11	Soil probe: 3-rod stainless FDR, RS485, VWC + temperature + EC, …
12	Front scan beam with drop-down under-leaf camera heads (v1 arms,…
13	2D lidar on a front mast
14	GNSS antenna (RTK-ready)
15	E-stop, latching, cuts the drive contactor
16	Dock: charging contacts, probe wash cup with brush, reference bl…

Mass 22.4 kg · weight 220 N

Centre of mass 0.9 mm from the probe axis, 186 mm above ground

Phantom lines: probe at 60 mm depth (stroke 290 mm)

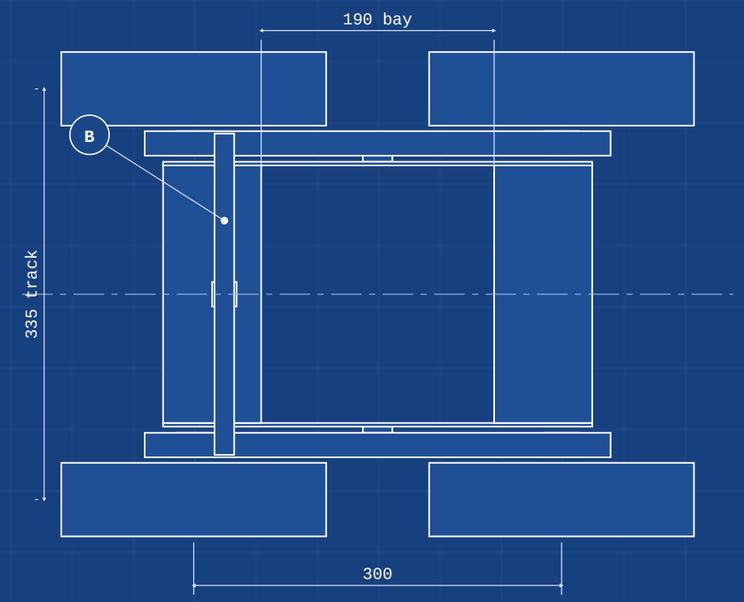
Open tool bay between the axles: the whole weight resists the push

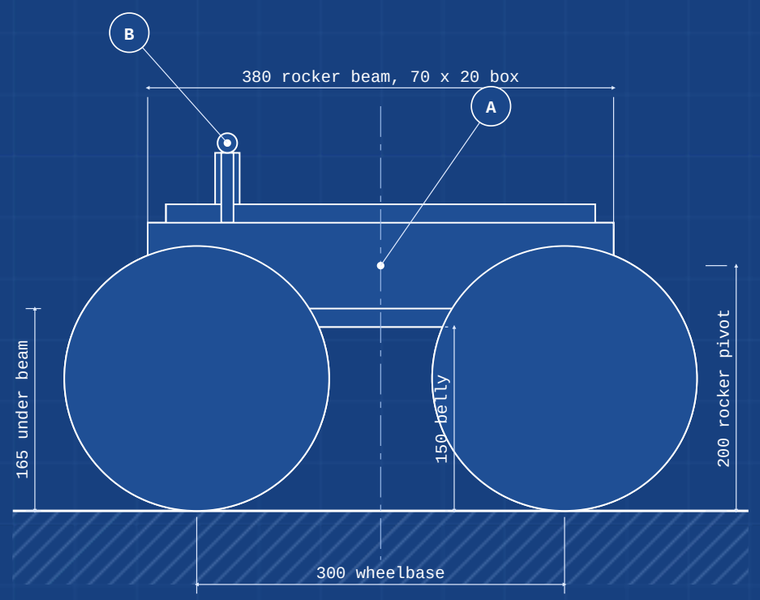
AGRIROVER v2

SOIL-MOISTURE SURVEY ROVER
design targets, not as-built

TITLE		SHEET
GENERAL ARRANGEMENT		**1 OF 5**
SCALE · UNITS · PROJECTION	SOURCE	TEAM · DATE
1:5 · mm · 3rd angle	**model.mjs**	**Auenchanters · 2026-11-11**

CHASSIS, PLAN





CHASSIS, LEFT SIDE

HOW THE ROCKER-DIFFERENTIAL WORKS

A Each side's two wheels hang from one rocker beam that pivots on the body.
B A differential bar across the front box links the beams: when one rises the other falls, and the body takes the average of their angles.
No springs: all four wheels stay loaded, and the body does not rise when the probe pushes down, so the push and the depth repeat.

STABILITY AND TERRAIN	VALUE	NOTE
Side slope before tipping (static)	42 deg	track / CG height
Fore-aft slope before tipping	39 deg	
100 mm clod under one wheel: beam tilt	19 deg	body about half
Body pitch on that clod	10 deg	
Braking needed to tip forward	0.80 g	stop is 0.2 g
Step climbed reliably	54 mm	half wheel radius [A]
Steepest grade	24 deg	mu 0.55 [A]
Pull available / rolling drag	121 / 22 N	soft tilled soil

MASS BUDGET [A]	kg	x	z	
hub-motor wheels, 4	9.6	0	108	
rocker beams and axle brackets, 2	1.3	0	200	
body: side plates, front and rear boxes	2.2	0	200	
differential bar, links, pivots	0.6	0	250	
battery 12S LiFePO4 6 Ah with BMS, in the rear cross box	2.4	-150	200	
2 dual BLDC controllers, front cross box	0.7	150	200	
probe module (rails, sled, linear axis and bellows, st…	1.9	0	400	
cleaning: station, pump, valves, water tank (full)	1.0	120	205	
electronics enclosure, Pi 5, MCU, DC-DC	0.9	-135	295	
lidar, cameras, GNSS, e-stop	0.5	0	370	
scan beam, arms and camera heads	0.8	295	170	
cables, connectors, fasteners	0.5	0	230	
TOTAL	22.4	-0.9	186	= centre of mass

MATERIALS

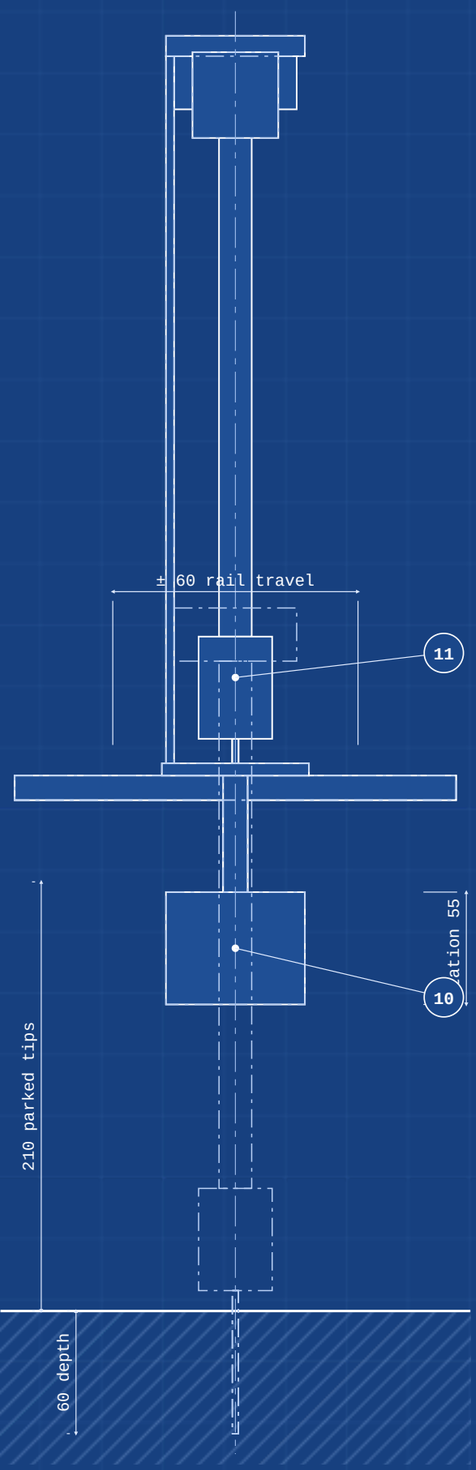
Rocker beams: aluminium box section with welded axle drop brackets.
Body: two 3 mm aluminium side plates joined by the front and rear cross boxes.
Pivots: stainless stub shafts in sealed bearings. Links: rod ends.
Wheels: 8.5 in pneumatic hub motors at about 60 kPa, sealant-filled.
[A] = assumption or vendor-class figure, to confirm on the bench.

AGRIROVER v2

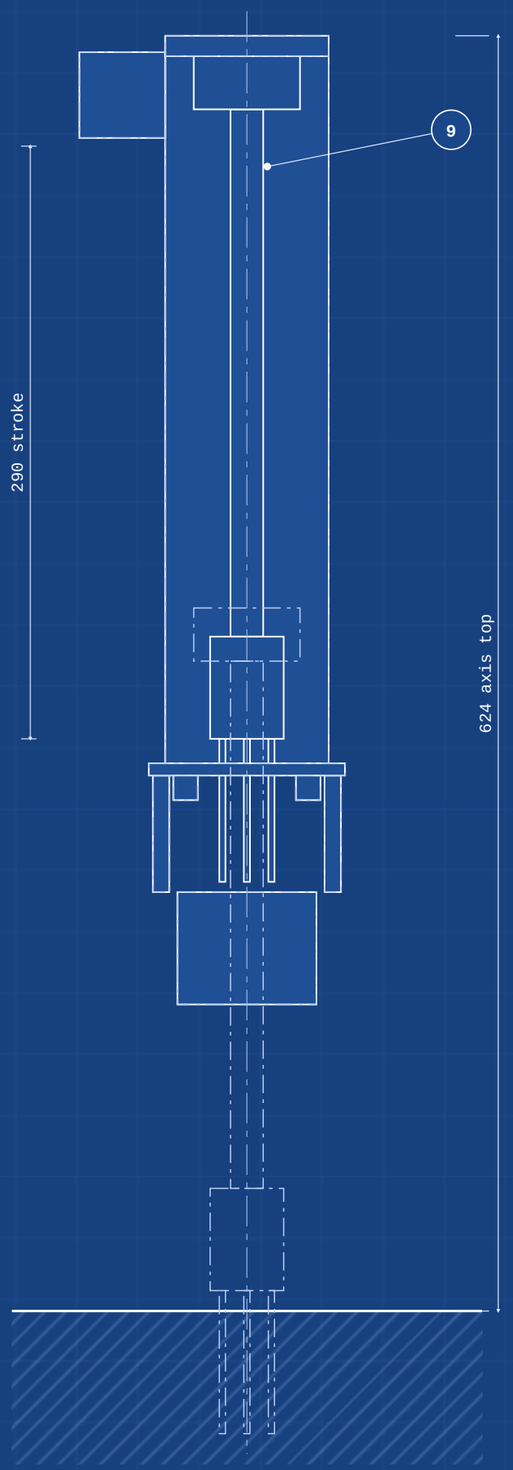
SOIL-MOISTURE SURVEY ROVER
design targets, not as-built

TITLE		SHEET
CHASSIS STRUCTURE		**2 OF 5**
SCALE · UNITS · PROJECTION	SOURCE	TEAM · DATE
1:5 · mm · 3rd angle	**model.mjs**	**Auenchanters · 2026-11-11**

PROBE MODULE, FRONT



PROBE MODULE, SIDE



PROBE FORCE	MOIST SOIL	DRY CRUST	
Push needed [A]	42 N	106 N	cone 1.2 / 3 MPa
Push available, probe centred	175 N	175 N	80% of lift-off
Push available, 60 mm off centre	129 N	129 N	
Margin over dry crust, centred		1.6 x	

t, s	PROBE CYCLE (one reading)	SPEED	
0 to 5.3	Lower to the surface	40 mm/s	
to 15.3	Insert 60 mm	6 mm/s	
to 17.3	Read (VWC, temperature, EC)	2 s	
to 24.0	Retract through wiper plate and brushes	40 mm/s	
to 27.0	Air knife	3 s	
to 28.0	Air check: reads clean-air baseline	1 s	within 0.02
+ 10	Wet clean when the air check fails twice	2 mL	about 10% of cycles [A]

LEVEL	WHEN	CLEANING, NO PERSON NEEDED
0	every cycle	PU wiper plate with three rod slots, then two strip brushes
1	every cycle	3 s air knife, then the air check
2	check fails	stroke through the wipers again, blast, check
3	still failing	1 s water spray, 8 s of air, check
4	after 2 wet cleans	reading marked invalid; survey goes on; rover rechecks later
5	at the dock	wash cup with brush, rinse, dry; reference block within 0.015

DROP-IN MODULE

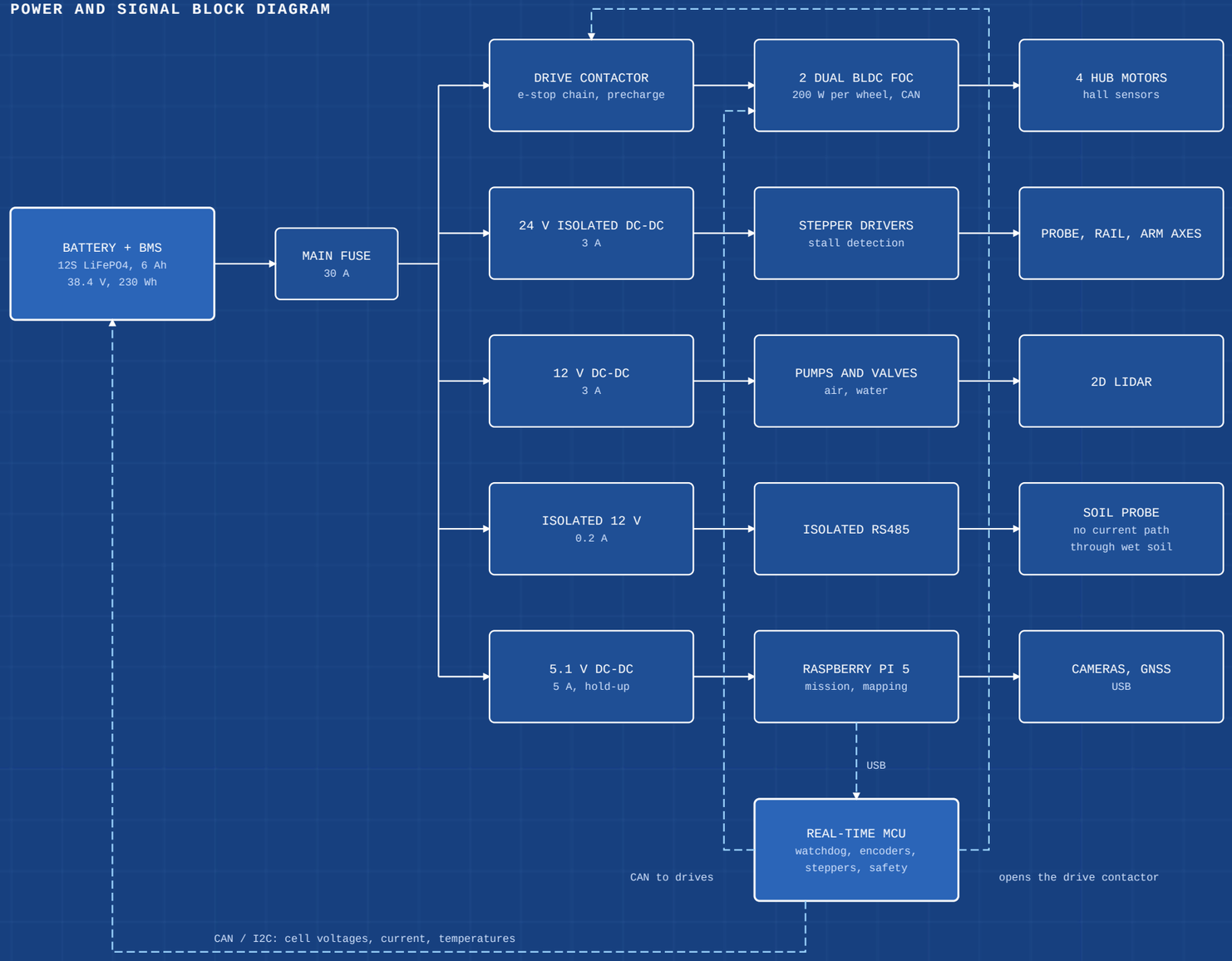
Two sideways rails, a sled, a belt-driven linear axis behind a bellows with its stepper beside it, the cleaning station hanging under the sled, and the probe. Held by two thumb screws, one M12 connector and one push-fit air/water line. The rail shifts the probe 5 cm to step round a stone, without driving. Probe: 3-rod stainless FDR, RS485, galvanically isolated from the rover. Phantom lines: full stroke on level ground.

AGRIROVER v2

SOIL-MOISTURE SURVEY ROVER
design targets, not as-built

TITLE		SHEET
PROBE MODULE AND CLEANING		**3 OF 5**
SCALE · UNITS · PROJECTION	SOURCE	TEAM · DATE
1:3 · mm · 3rd angle	**model.mjs**	**Auenchanters · 2026-11-11**

POWER AND SIGNAL BLOCK DIAGRAM



BMS LIMITS	SETTING
Cell over / under voltage	3.65 / 2.5 V
Soft cut-off (rover returns to dock)	2.9 V per cell
Continuous / peak current	25 / 50 A
Short circuit	120 A in 200 us
Charge temperature	0 to 45 C
Discharge temperature	-20 to 60 C
Balancing	> 30 mV, 60 mA
Charge current at the dock	3 A (0.5 C)
Return to dock at	30% charge

ENERGY FOR ONE SURVEY	Wh
Driving 466 m	5.2
32 turns in place	4.8
Electronics, always on	10.8
Probe axis, 60 readings	4.8
Cleaning	0.5
TOTAL	26.1
Usable in the pack (80%)	184
Surveys per charge	7.1
Mission time	46 min

SUPPLY RAIL	FEEDS
Drive	2 dual BLDC FOC controllers, current-limited
24 V, 3 A	probe stepper, Y-rail stepper, scan-arm steppers
12 V, 3 A	air pump, water pump, solenoid valves, lidar
5.1 V, 5 A	Raspberry Pi 5 and USB cameras
Probe, 12 V, 0.2 A	the soil probe: no current path through wet soil

WHY LiFePO4

Far harder to push into thermal runaway than nickel-rich Li-ion, about 2,000 cycles [A], and its extra weight sits low and over the probe, where it adds push.

SAFETY DOES NOT DEPEND ON SOFTWARE

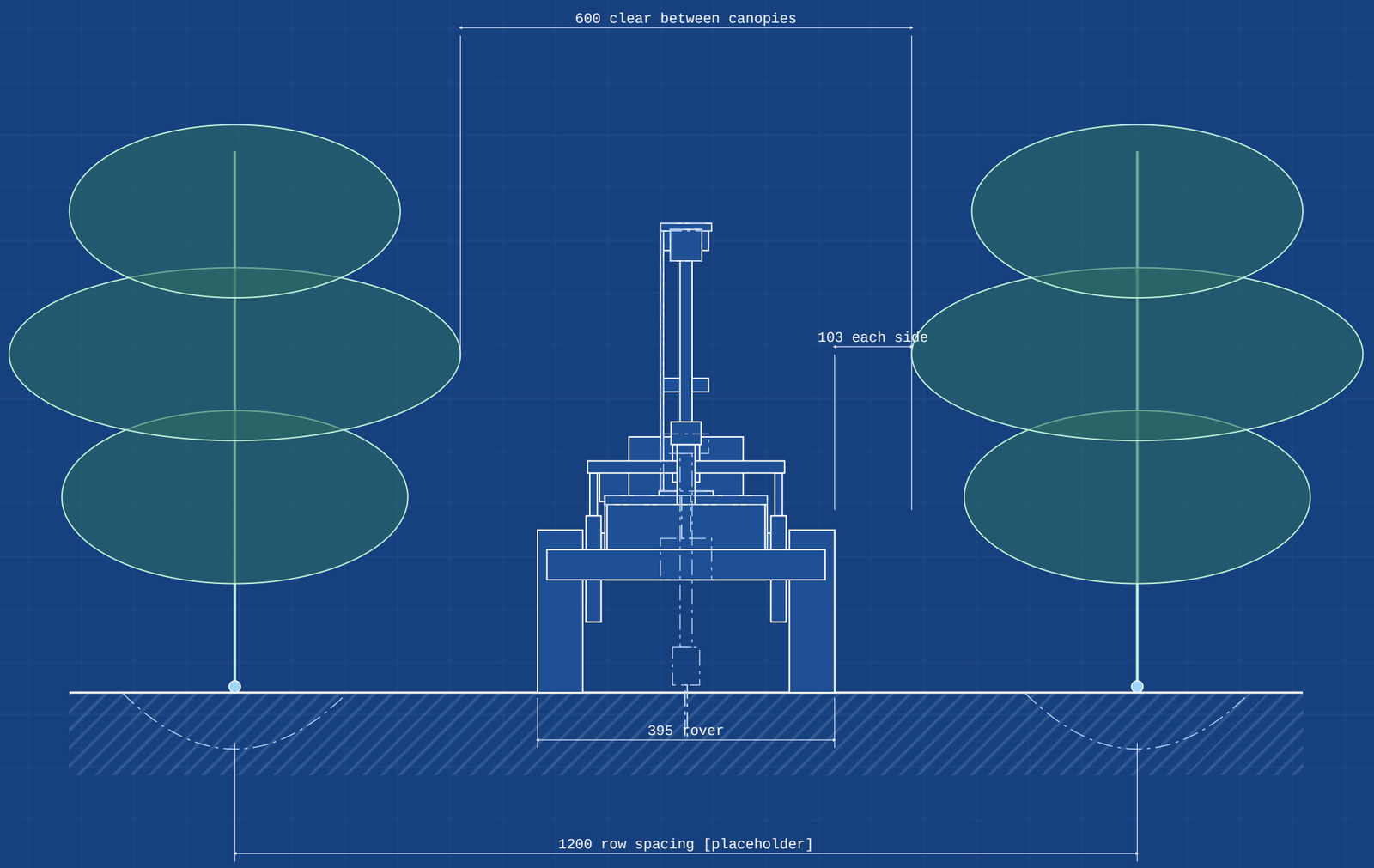
The e-stop and the RC kill switch open the drive contactor directly.
The MCU stops the motors if the Pi's heartbeat is missing for 200 ms.
The motor controllers stop if the MCU's CAN messages stop.
IP65 enclosure, M12/M8 IP67 connectors, membrane vent, conformal-coated boards.
Solid line: power. Dashed line: signal.

AGRIROVER v2

SOIL-MOISTURE SURVEY ROVER
design targets, not as-built

TITLE		SHEET
ELECTRICAL AND POWER		**4 OF 5**
SCALE · UNITS · PROJECTION	SOURCE	TEAM · DATE
NTS · mm · 3rd angle	**model.mjs**	**Auenchanters · 2026-11-11**

ROVER IN THE CROP LANE, SEEN FROM THE FRONT



WHAT THE PROBE CAN AND CANNOT SEE

The rover drives the lane centre and probes there, 600 mm from each drip line (blue dot).
The wetted strip under a drip line (dashed) is about 150 mm to each side: the probe's
± 60 mm rail does not reach it. A reach of 300 mm would cover half of it.
Simulated with the new soil physics: a drip run during the survey raises the map error,
because the water lands where lane-centre probes never go (docs/CALIBRATION.md).
Open design question: a longer side reach, or a second probe on an arm.

SERVICE SCHEDULE [A]

WEAR PART	LIFE	UNIT
Wiper plate (polyurethane)	5,000	probe cycles
Strip brushes	10,000	probe cycles
Pump inlet filter	20,000	probe cycles
Belt and carriage bearings (behind…	50,000	probe cycles
Probe (rods, seal)	30,000	probe cycles
Tyres	300	km
Battery (LiFePO4)	2,000	full cycles to 80…
Dock water reservoir (10 L)	5	days of 3 surveys

SOURCING

REF	BUY OR MAKE
1	buy
2	make
3	make + buy
4	make
5	make + buy
6	make + buy
7	make
8	buy + make
9	make + buy
10	make + buy
11	buy
12	make
13	buy
14	buy
15	buy
16	make + buy

OPEN QUESTIONS BEFORE BUILDING

1 Soil strength on the target farms (sets the push).
2 How often each cleaning level is needed in local clay.
3 The reference block for the bias check.
4 Position along the row without RTK.
5 Rocker stiffness and canopy clearance in a real lane.
6 Side reach toward the drip line (this sheet).

AGRIROVER v2

SOIL-MOISTURE SURVEY ROVER
design targets, not as-built

TITLE		SHEET
FIELD ENVELOPE AND SERVICE		**5 OF 5**
SCALE · UNITS · PROJECTION	SOURCE	TEAM · DATE
1:7 · mm · 3rd angle	**model.mjs**	**Auenchanters · 2026-11-11**